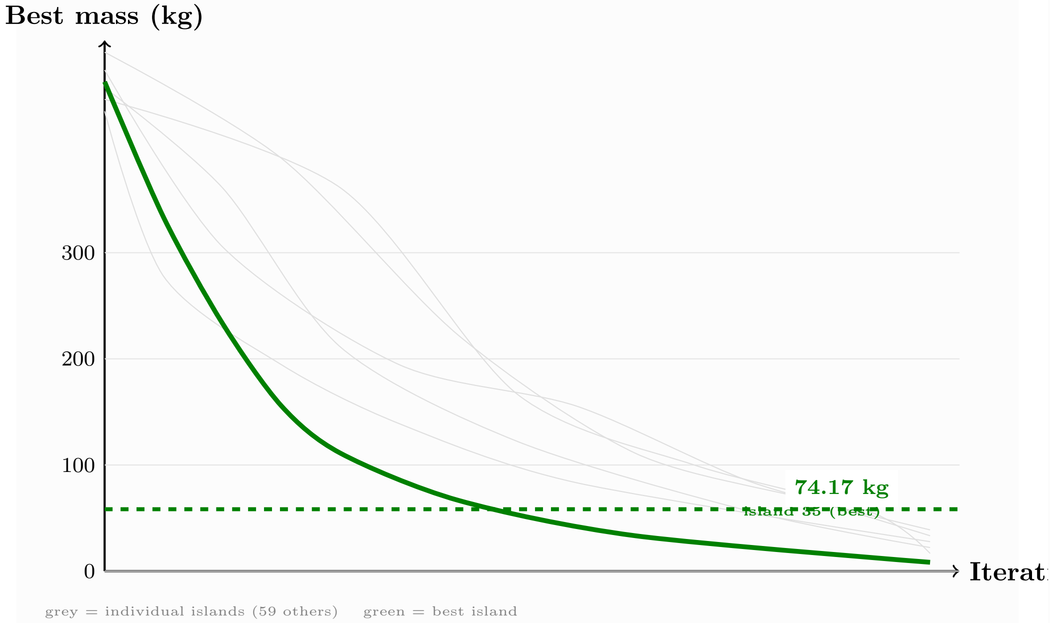
The V6.2 Optimization Landscape (Corrected Physics)

Three corrections (tan→sin, $\cos^3 \to \cos^{2.65}$, coupled knuckles) shifted the optimum to 74.17 kg at $n = 12$

1. Convergence: 60 Islands × 10 000 Steps

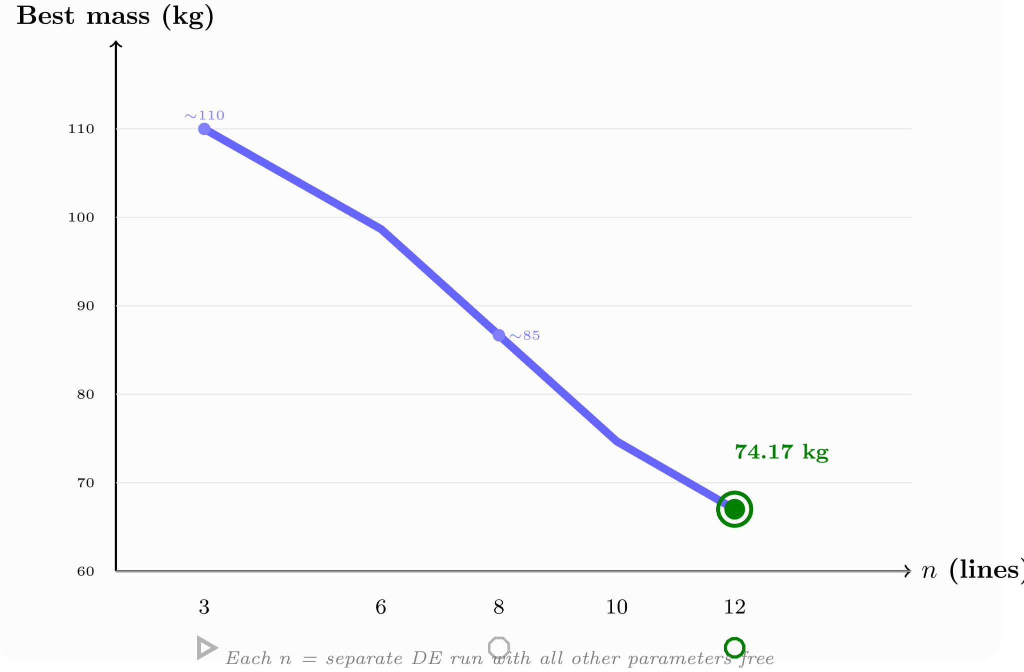
Each grey trace = one island population. Green = best island (global best).



Bounds: $n \in [3, 12]$, $r_{hub} \in [1, 10]$ m, $\beta \in [-0.8, 0.8]$, $n_{rings} \in [5, 16]$, $n_{exp} \in [0, 6]$, 600k evals

2. Polygon Effect: n_{lines} vs Mass

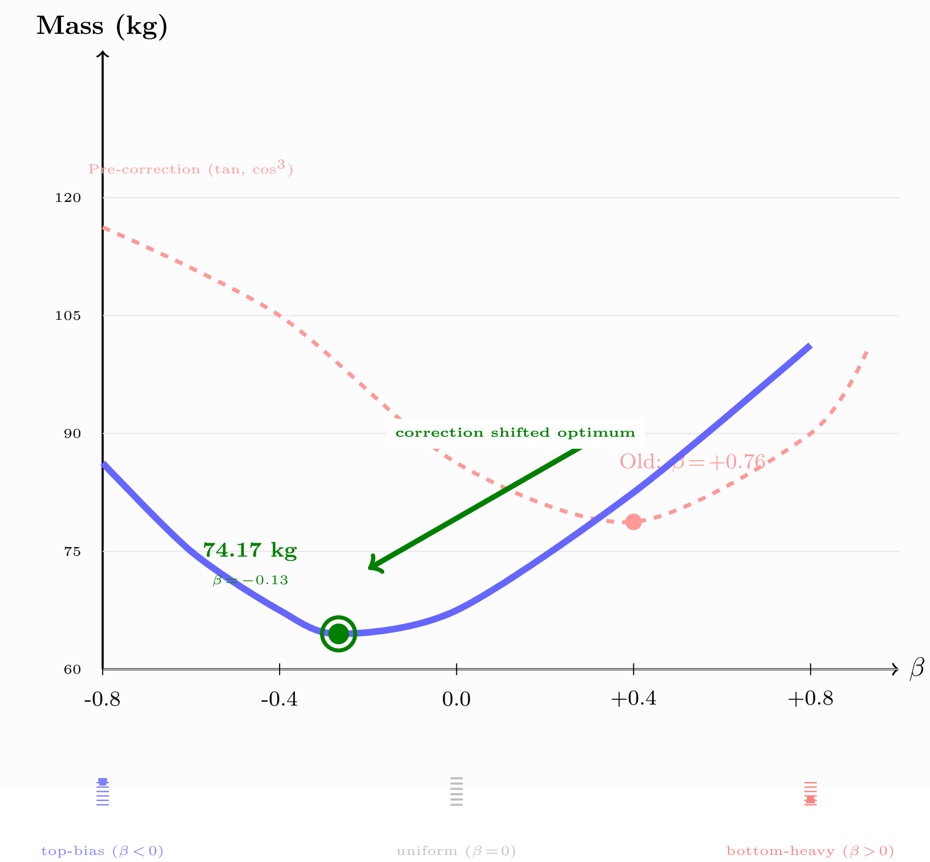
More lines = thinner beams = smaller knuckles = lower total mass



Each n = separate DE run with all other parameters free

3. Density Profile β: Sign Flip from +0.76 to −0.13

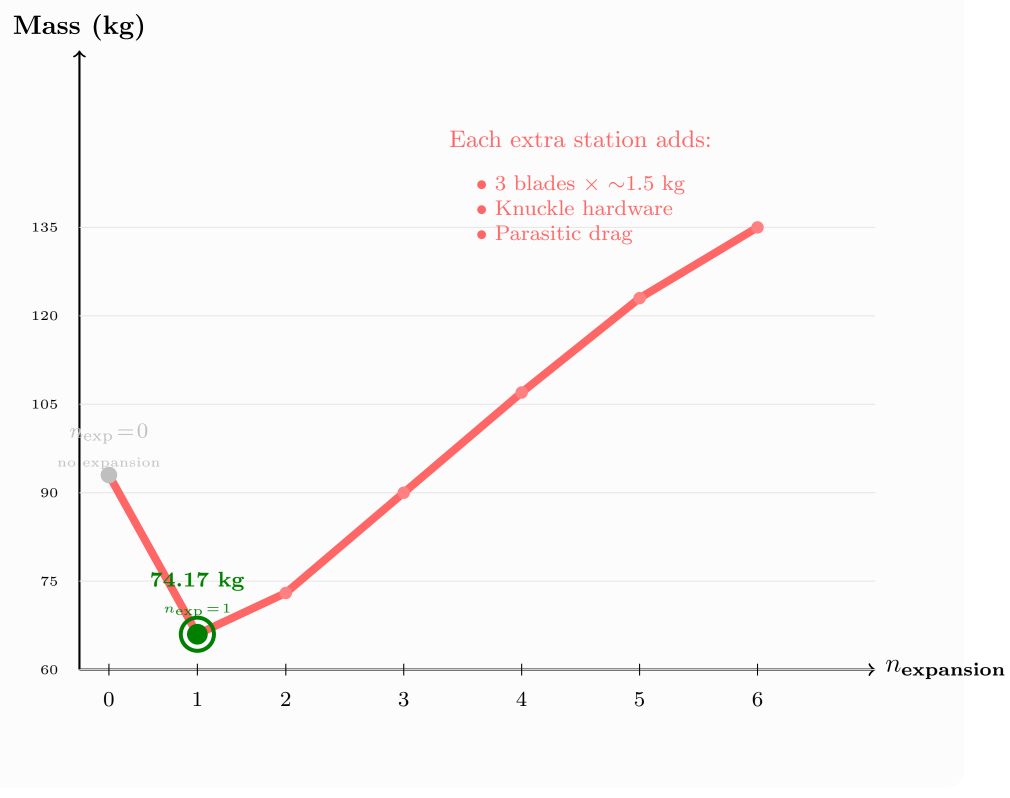
$\beta > 0$ = bottom-heavy (rings cluster low). $\beta < 0$ = top-bias. $\beta = 0$ = uniform spacing.



The physics correction (tan→sin + $\cos^{2.65}$) halved the buckling demand, making top-bias optimal

4. Expansion Stations: n_{exp} vs Mass

Expansion rotors sit at ring positions on the shaft (NOT at main rotor). More stations = more mass.



$n_{exp} = 0$ (no expansion) is heavier than $n_{exp} = 1$ — one expansion station at the hub ring provides enough F_r